

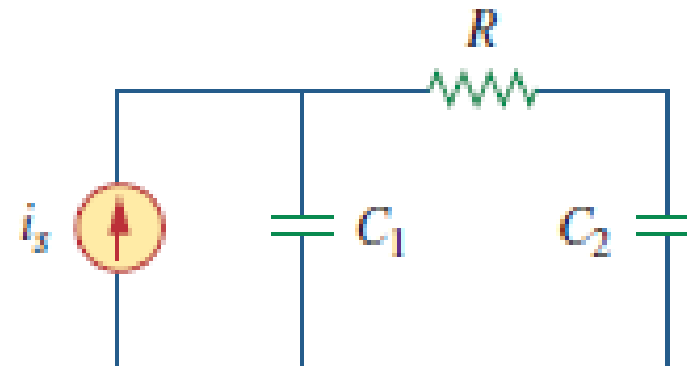
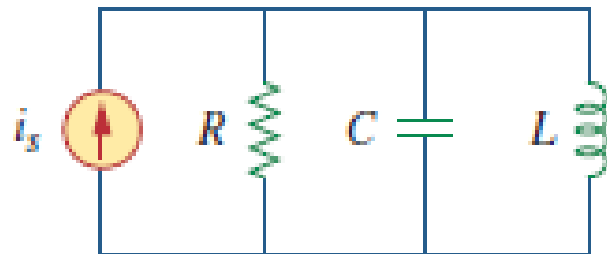
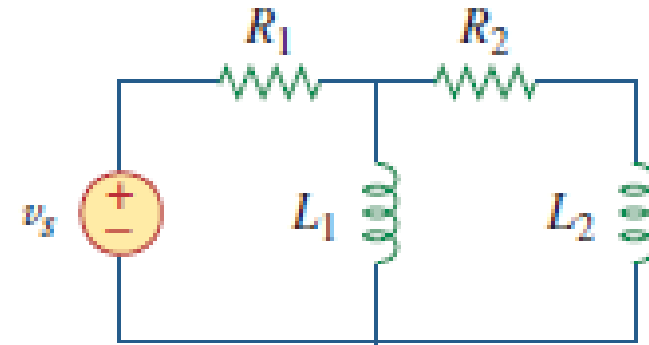
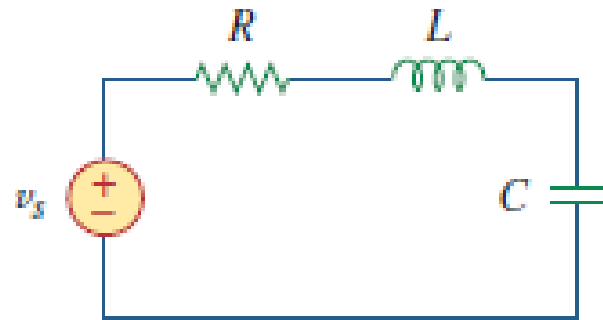
Module -2

Source Free Response of RLC Circuits

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Second order Circuits

A **second-order circuit** is characterized by a second-order differential equation. It consists of resistors and the equivalent of two energy storage elements.



Source Free Series RLC Circuit

The circuit is being excited by the energy initially stored in the capacitor and inductor.

The energy is represented by the initial capacitor voltage V_0 and initial inductor current I_0 .

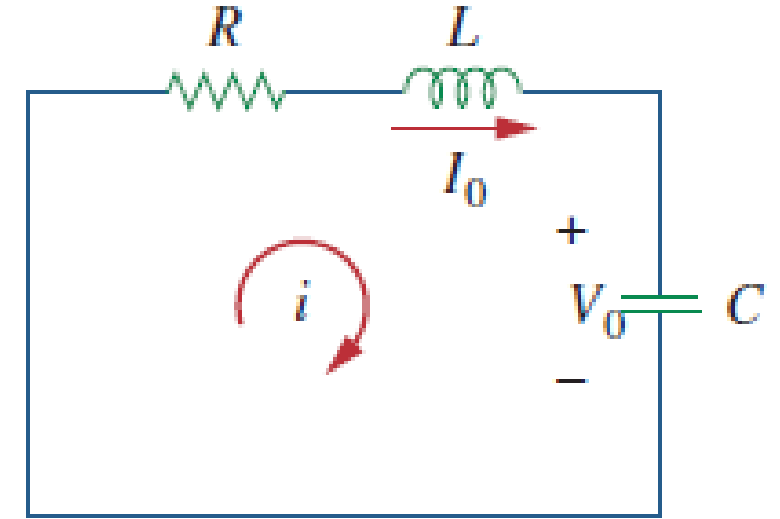
At $t = 0$,

$$v(0) = \frac{1}{C} \int_{-\infty}^0 i dt = V_0 \longrightarrow (1)$$

$$i(0) = I_0 \longrightarrow (2)$$

Applying KVL around the loop,

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = 0 \longrightarrow (3)$$



To eliminate the integral, we differentiate with respect to t and rearrange terms. We get

$$\frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0 \quad \longrightarrow (4)$$

This is a *second-order differential equation*

To solve such a second-order differential equation requires that we have two initial conditions, such as the initial value of i and its first derivative or initial values of some i and v .

From Eqn. (1) and (3),

$$Ri(0) + L \frac{di(0)}{dt} + V_0 = 0$$

$$\frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0) \quad \longrightarrow (5)$$

$$i = Ae^{st} \longrightarrow (6)$$

Using eqn. (6) in (4),

$$As^2e^{st} + \frac{AR}{L}se^{st} + \frac{A}{LC}e^{st} = 0$$

$$Ae^{st}\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right) = 0$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \longrightarrow (7)$$

This quadratic equation is known as the *characteristic equation*

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \longrightarrow (8)$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

where

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

S_1, S_2 – Natural frequencies (nepers per second)

ω_0 – undamped natural frequency / resonant frequency (radians per second)

α – neper frequency / damping factor (nepers per second)

$$s^2 + 2\alpha s + \omega_0^2 = 0$$

$$i_1 = A_1 e^{s_1 t}, \quad i_2 = A_2 e^{s_2 t}$$

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

where the constants A_1 and A_2 are determined from the initial values $i(0)$ and $di(0)/dt$

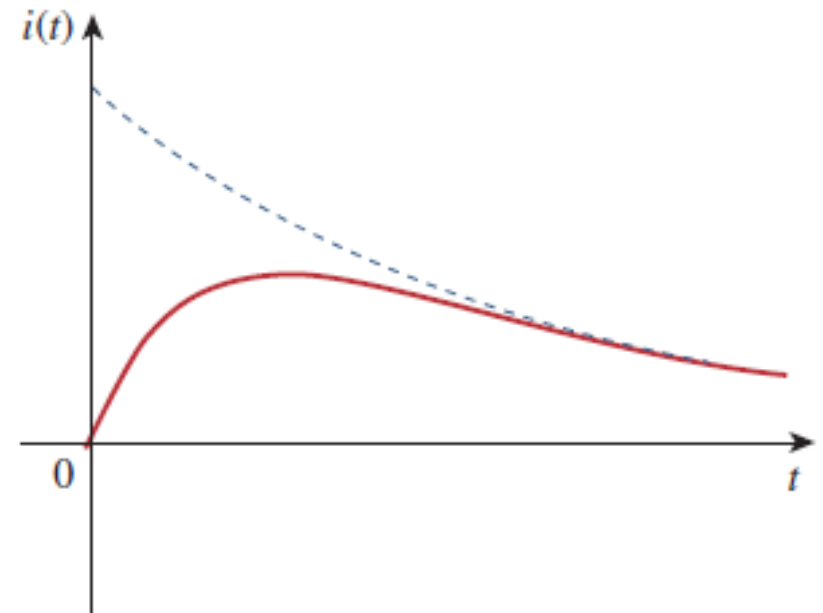
- Three types of solutions:
 1. If $\alpha > \omega_0$, we have the *overdamped* case.
 2. If $\alpha = \omega_0$, we have the *critically damped* case.
 3. If $\alpha < \omega_0$, we have the *underdamped* case.

Overdamped Case ($\alpha > \omega_0$)

$\alpha > \omega_0$ implies $C > 4L/R^2$.

both roots s_1 and s_2 are negative and real.

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

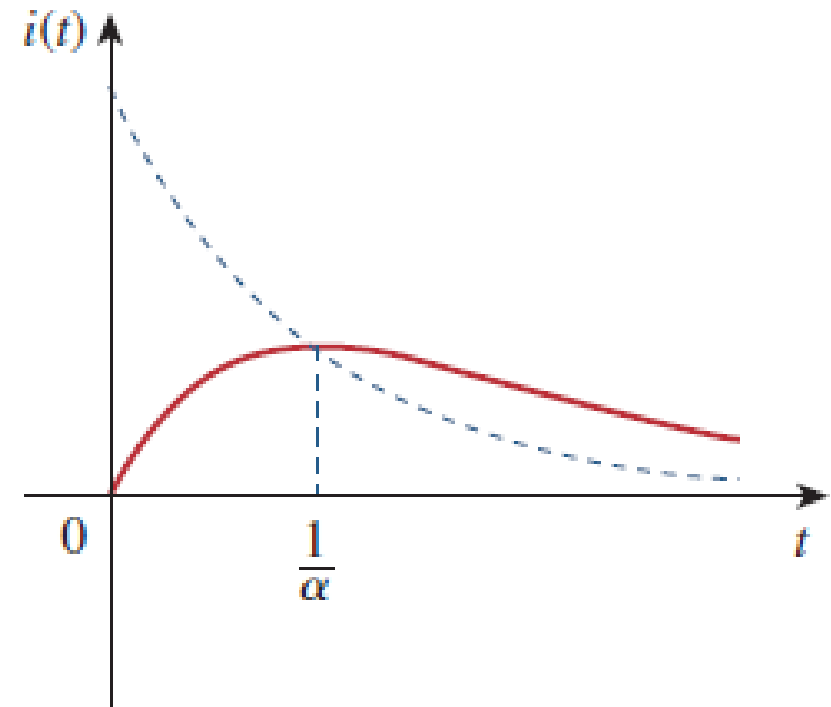


Critically Damped Case ($\alpha = \omega_0$)

When $\alpha = \omega_0$, $C = 4L/R^2$ and

$$s_1 = s_2 = -\alpha = -\frac{R}{2L}$$

$$i(t) = (A_2 + A_1 t)e^{-\alpha t}$$



Underdamped Case ($\alpha < \omega_0$)

For $\alpha < \omega_0$, $C < 4L/R^2$.

$$s_1 = -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha + j\omega_d$$

$$s_2 = -\alpha - \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha - j\omega_d$$

where $j = \sqrt{-1}$ and $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$,

ω_d is called the *damped natural frequency*.

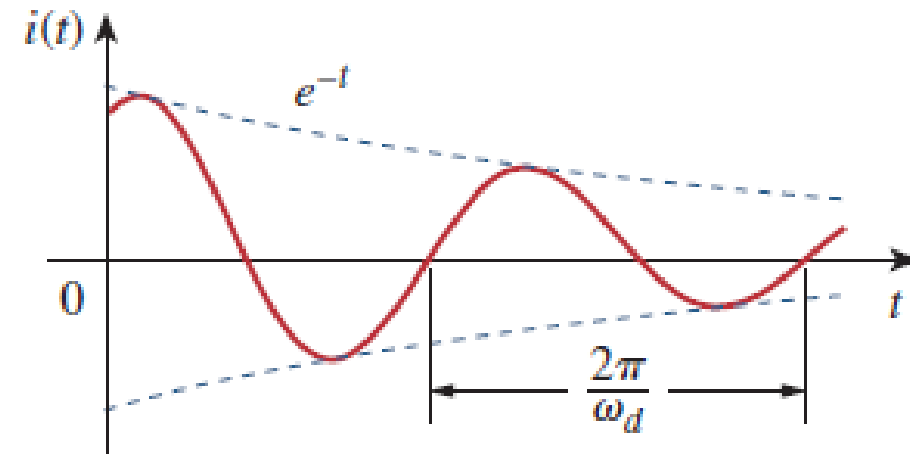
$$\begin{aligned} i(t) &= A_1 e^{-(\alpha - j\omega_d)t} + A_2 e^{-(\alpha + j\omega_d)t} \\ &= e^{-\alpha t} (A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t}) \end{aligned}$$

Using Euler's identities,

$$\begin{aligned} i(t) &= e^{-\alpha t} [A_1(\cos \omega_d t + j \sin \omega_d t) + A_2(\cos \omega_d t - j \sin \omega_d t)] \\ &= e^{-\alpha t} [(A_1 + A_2) \cos \omega_d t + j(A_1 - A_2) \sin \omega_d t] \end{aligned}$$

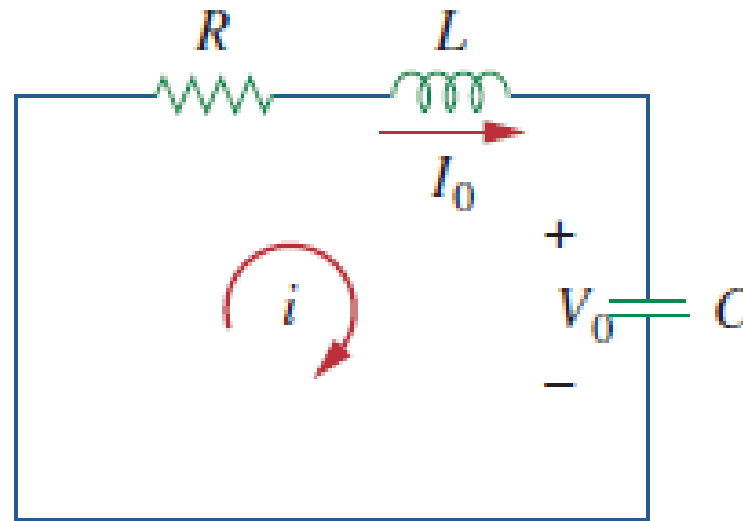
Replacing constants $(A_1 + A_2)$ and $j(A_1 - A_2)$ with constants B_1 and B_2 ,

$$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$



Numerical - 1

Calculate the characteristic roots of the circuit. Is the natural response overdamped, under damped, or critically damped? Given $R = 40\ \Omega$, $L = 4\ \text{H}$, and $C = \frac{1}{4}\ \text{F}$.



Solution:

We first calculate

$$\alpha = \frac{R}{2L} = \frac{40}{2(4)} = 5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times \frac{1}{4}}} = 1$$

The roots are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -5 \pm \sqrt{25 - 1}$$

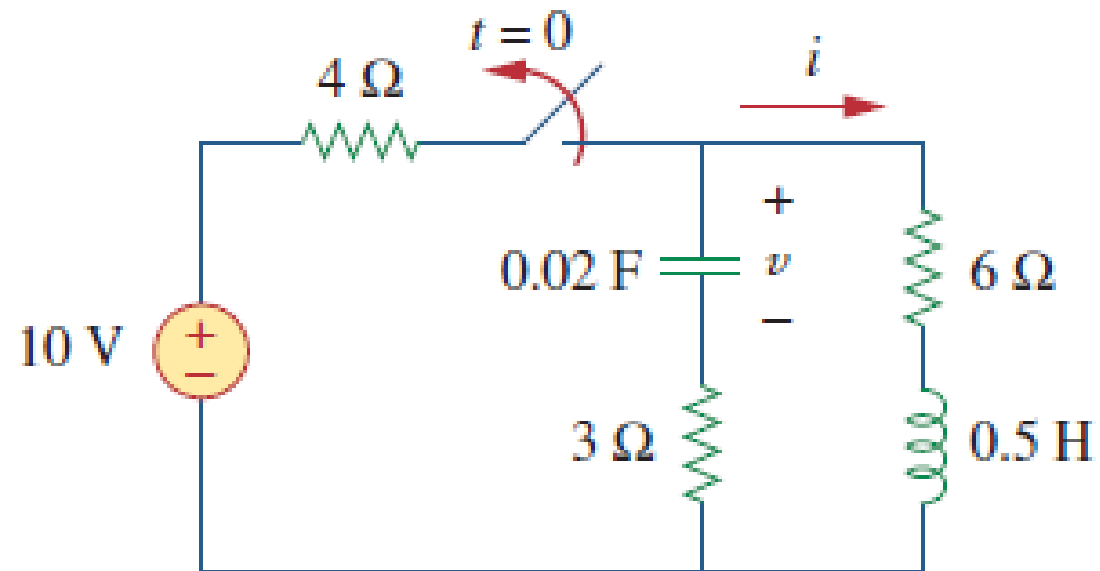
or

$$s_1 = -0.101, \quad s_2 = -9.899$$

Since $\alpha > \omega_0$, we conclude that the response is overdamped. This is also evident from the fact that the roots are real and negative.

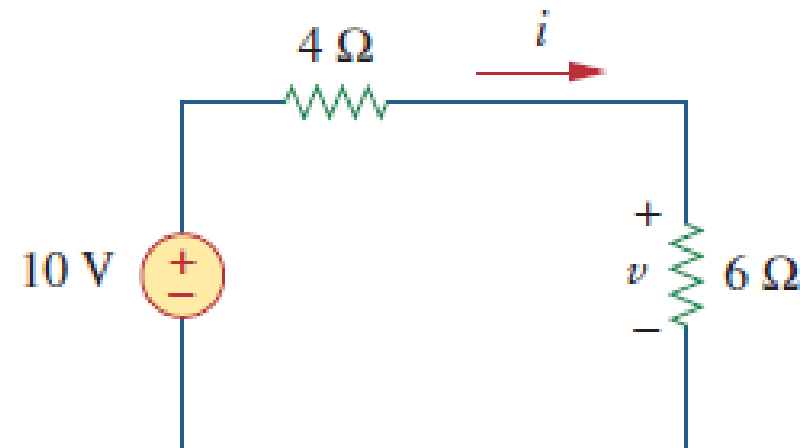
Numerical - 2

Find $i(t)$ in the circuit. Assume that the circuit has reached steady state at $t = 0^-$.



Solution:

For $t < 0$, the switch is closed. The capacitor acts like an open circuit while the inductor acts like a shunted circuit. The equivalent circuit is



$$i(0) = \frac{10}{4 + 6} = 1 \text{ A},$$

$$v(0) = 6i(0) = 6 \text{ V}$$

For $t > 0$, the switch is opened

$$\alpha = \frac{R}{2L} = \frac{9}{2(\frac{1}{2})} = 9,$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{2} \times \frac{1}{50}}} = 10$$

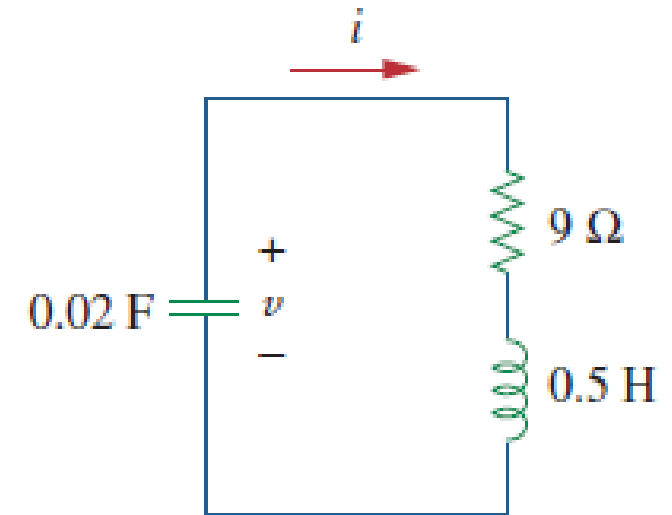
$$s_{1,2} = -9 \pm j4.359$$

Hence, the response is underdamped ($\alpha < \omega$); that is,

$$i(t) = e^{-9t}(A_1 \cos 4.359t + A_2 \sin 4.359t)$$

We now obtain A_1 and A_2 using the initial conditions. At $t = 0$,

$$i(0) = 1 = A_1$$



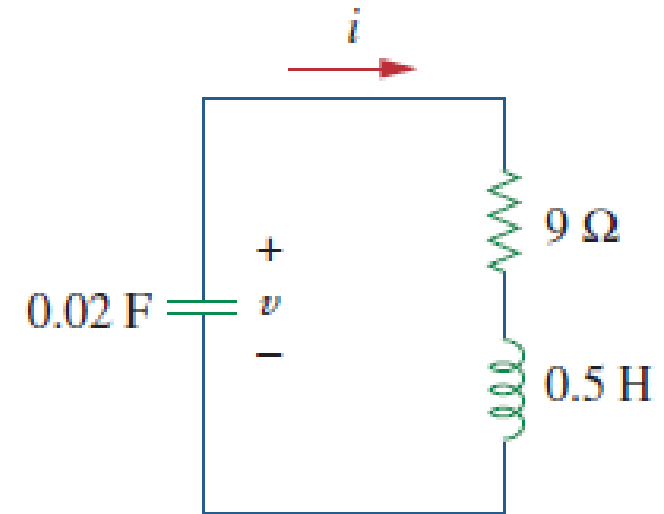
- Applying KVL to the loop,

$$9 i(0) + 0.5 \, di(0)/dt - v(0) = 0$$

$$di(0)/dt = \{-9i(0) + v(0)\} / 0.5$$

$$di(0)/dt = \{-9 (1) + 6\} / 0.5$$

$$di(0)/dt = -6 \, \text{A/s}$$



$$\begin{aligned} \frac{di}{dt} = & -9e^{-9t}(A_1 \cos 4.359t + A_2 \sin 4.359t) \\ & + e^{-9t}(4.359)(-A_1 \sin 4.359t + A_2 \cos 4.359t) \end{aligned}$$

at $t = 0$ gives

$$-6 = -9(A_1 + 0) + 4.359(-0 + A_2)$$

But $A_1 = 1$

$$-6 = -9 + 4.359A_2 \quad \Rightarrow \quad A_2 = 0.6882$$

$$i(t) = e^{-9t}(\cos 4.359t + 0.6882 \sin 4.359t) \text{ A}$$

Source Free Parallel RLC Circuit

The circuit is being excited by the energy initially stored in the capacitor and inductor.

The energy is represented by the initial capacitor voltage V_0 and initial inductor current I_0 .

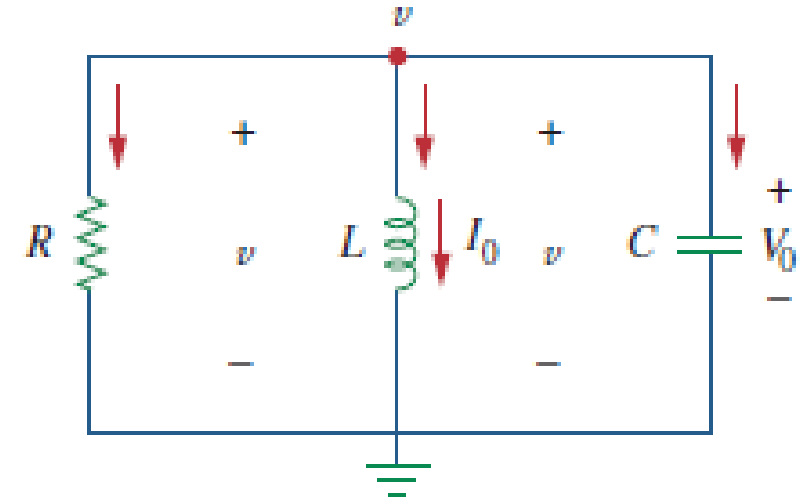
At $t = 0$,

$$i(0) = I_0 = \frac{1}{L} \int_{-\infty}^0 v(t) dt \longrightarrow (1)$$

$$v(0) = V_0 \longrightarrow (2)$$

Applying KCL at the top node,

$$\frac{v}{R} + \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau + C \frac{dv}{dt} = 0 \longrightarrow (3)$$



To eliminate the integral, we differentiate with respect to t and rearrange terms. We get

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0 \longrightarrow (4)$$

This is a *second-order differential equation*

We obtain the characteristic equation by replacing the first derivative by s and the the second derivative by s^2 .

$$s^2 + \frac{1}{RC} s + \frac{1}{LC} = 0 \longrightarrow (5)$$

The roots of the characteristic equation are

$$s_{1,2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \longrightarrow (6)$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \longrightarrow (7)$$

$$\alpha = \frac{1}{2RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}} \longrightarrow (8)$$

Overdamped Case ($\alpha > \omega_0$)

$$\alpha > \omega_0 \text{ when } L > 4R^2C.$$

Roots are real and negative

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Critically Damped Case ($\alpha = \omega_0$)

$\alpha = \omega_0, L = 4R^2C.$ The roots are real and equal

$$v(t) = (A_1 + A_2 t)e^{-\alpha t}$$

Underdamped Case ($\alpha < \omega_0$)

$\alpha < \omega_0, L < 4R^2C.$ the roots are complex

$$s_{1,2} = -\alpha \pm j\omega_d \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$v(t) = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

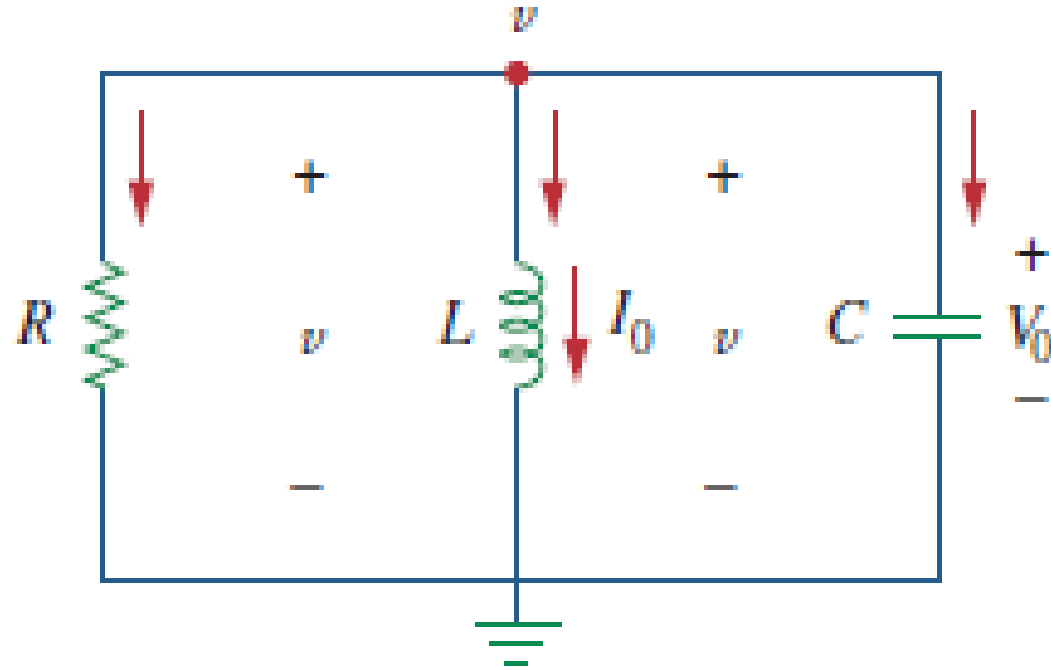
- To find the initial condition,
- From Eqn. (1) and (3),

$$\frac{V_0}{R} + I_0 + C \frac{dv(0)}{dt} = 0$$

$$\frac{dv(0)}{dt} = -\frac{(V_0 + RI_0)}{RC}$$

Numerical - 1

Find $v(t)$ in the circuit for $t > 0$. Assuming $v(0) = 5$ V, $i(0) = 0$, $L = 1$ H, $C = 10$ mF. Consider these cases: $R = 1.923$ Ω , $R = 5$ Ω , and $R = 6.25$ Ω .



Solution:

■ **CASE 1** If $R = 1.923 \Omega$,

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 1.923 \times 10 \times 10^{-3}} = 26$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10 \times 10^{-3}}} = 10$$

Since $\alpha > \omega_0$ in this case, the response is overdamped. The roots of the characteristic equation are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -2, -50$$

and the corresponding response is

$$v(t) = A_1 e^{-2t} + A_2 e^{-50t}$$

We now apply the initial conditions to get A_1 and A_2 .

$$v(0) = 5 = A_1 + A_2$$

$$\frac{dv(0)}{dt} = -\frac{v(0) + Ri(0)}{RC} = -\frac{5 + 0}{1.923 \times 10 \times 10^{-3}} = -260$$

$$\frac{dv}{dt} = -2A_1e^{-2t} - 50A_2e^{-50t}$$

At $t = 0$,

$$-260 = -2A_1 - 50A_2$$

$$A_1 = -0.2083 \text{ and } A_2 = 5.208.$$

$$v(t) = -0.2083e^{-2t} + 5.208e^{-50t}$$

■ **CASE 2** When $R = 5 \Omega$,

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 5 \times 10 \times 10^{-3}} = 10$$

while $\omega_0 = 10$ remains the same. Since $\alpha = \omega_0 = 10$, the response is critically damped. Hence, $s_1 = s_2 = -10$, and

$$v(t) = (A_1 + A_2 t)e^{-10t}$$

To get A_1 and A_2 , we apply the initial conditions

$$v(0) = 5 = A_1$$

$$\frac{dv(0)}{dt} = -\frac{v(0) + Ri(0)}{RC} = -\frac{5 + 0}{5 \times 10 \times 10^{-3}} = -100$$

$$\frac{dv}{dt} = (-10A_1 - 10A_2t + A_2)e^{-10t}$$

At $t = 0$,

$$-100 = -10A_1 + A_2$$

$$v(t) = (5 - 50t)e^{-10t} \text{ V}$$

■ **CASE 3** When $R = 6.25 \, \Omega$,

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 6.25 \times 10 \times 10^{-3}} = 8$$

while $\omega_0 = 10$ remains the same. As $\alpha < \omega_0$ in this case, the response is underdamped. The roots of the characteristic equation are

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -8 \pm j6$$

$$v(t) = (A_1 \cos 6t + A_2 \sin 6t)e^{-8t}$$

We now obtain A_1 and A_2 , as

$$v(0) = 5 = A_1$$

$$\frac{dv(0)}{dt} = -\frac{v(0) + Ri(0)}{RC} = -\frac{5 + 0}{6.25 \times 10 \times 10^{-3}} = -80$$

$$\frac{dv}{dt} = (-8A_1 \cos 6t - 8A_2 \sin 6t - 6A_1 \sin 6t + 6A_2 \cos 6t)e^{-8t}$$

$$\text{At } t = 0, \quad -80 = -8A_1 + 6A_2$$

$$v(t) = (5 \cos 6t - 6.667 \sin 6t)e^{-8t}$$